Proceedings of the International Conference on Internet of Things & Engineering · Published 2025

Collaborative Federated Edge Analytics for Predictive Fleet Maintenance in Connected IoT

Matteo Rossi, Dr. Chloe Bennett, Mateo Alvarez

Department of Computer Science & Intelligent Systems · Publication Year: 2025

ABSTRACT

Connected commercial vehicle fleets generate rich telemetry containing proprietary performance data that operators refuse to centralize. We implement a collaborative federated edge analytics framework that trains distributed vibration and thermal degradation predictors on vehicle telematics units. Vehicles train localized predictive models during idle charging windows, uploading only encrypted model deltas to cloud fleet servers. Across a deployed commercial fleet of 450 transit buses, the system anticipated gearbox and alternator component failures 18 days prior to breakdown. The unresolved challenge is managing straggler updates when vehicles traverse intermittent cellular coverage zones during multi-day rural transit routes.

Keywords— Internet of Things, Federated Analytics, Edge Computing, Predictive Maintenance

INTRODUCTION

Commercial vehicle downtime costs fleet operators thousands of dollars per vehicle per day. Fleet owners are reluctant to stream raw engine telemetry to third-party cloud vendors due to competitive concerns.

METHODOLOGY

Each bus operates an on-board edge computer running federated gradient descent on CAN-bus sensor streams. Federated averaging is conducted opportunistically whenever vehicles connect to depot high-speed WiFi networks.

RESULTS

Over a 12-month pilot across 450 municipal buses, the predictive system prevented 38 roadside breakdowns, reducing unplanned maintenance costs by $420,000 while maintaining 100% on-premise telemetry privacy.

LIMITATIONS

Vehicles operating rural multi-day routes frequently miss aggregation cycles, causing model parameter staleness and requiring asynchronous federated buffering protocols that consume on-board storage.

REFERENCES

[1] Dr. Ramesh Kulkarni, Matteo Rossi, Feng-Hua Lin (2021). "Decentralized Edge Intelligence in Resource-Constrained Industrial IoT Sensor Grids". In Proc. Int. Conf. on Advanced Computing Systems.

[2] Matteo Rossi, Dr. Chloe Bennett, Amara Diallo (2023). "Zero-Trust Device Identity and Behavioral Anomaly Detection in Smart City IoT Networks". In Proc. Int. Conf. on Advanced Computing Systems.

[3] Dr. Ramesh Kulkarni, Feng-Hua Lin, Mateo Alvarez (2024). "Adaptive Long-Range LoRaWAN Optimization for Real-Time Agricultural Sensor Telemetry". In Proc. Int. Conf. on Advanced Computing Systems.

[4] Vaswani, A., et al. (2017). "Attention Is All You Need". Advances in Neural Information Processing Systems, 30.

[5] Kipf, T. N., & Welling, M. (2016). "Semi-Supervised Classification with Graph Convolutional Networks". arXiv:1609.02907.

[6] Lewis, P., et al. (2020). "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks". NeurIPS.